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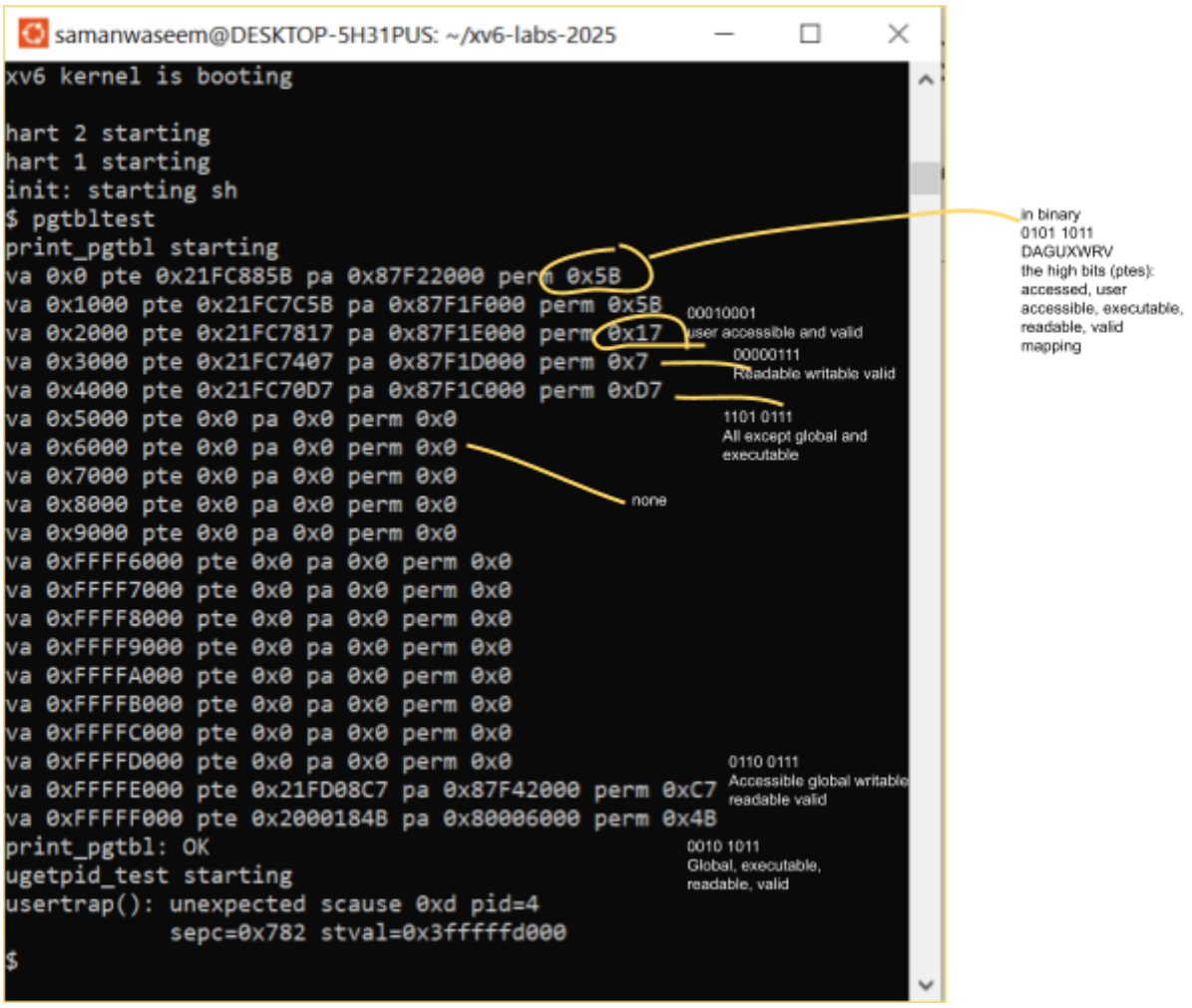
1. Inspect a user-process page table



Reference:

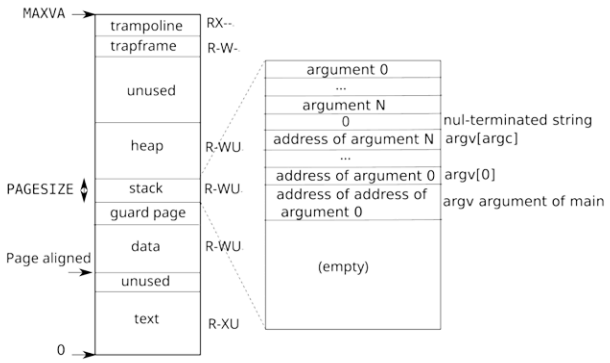
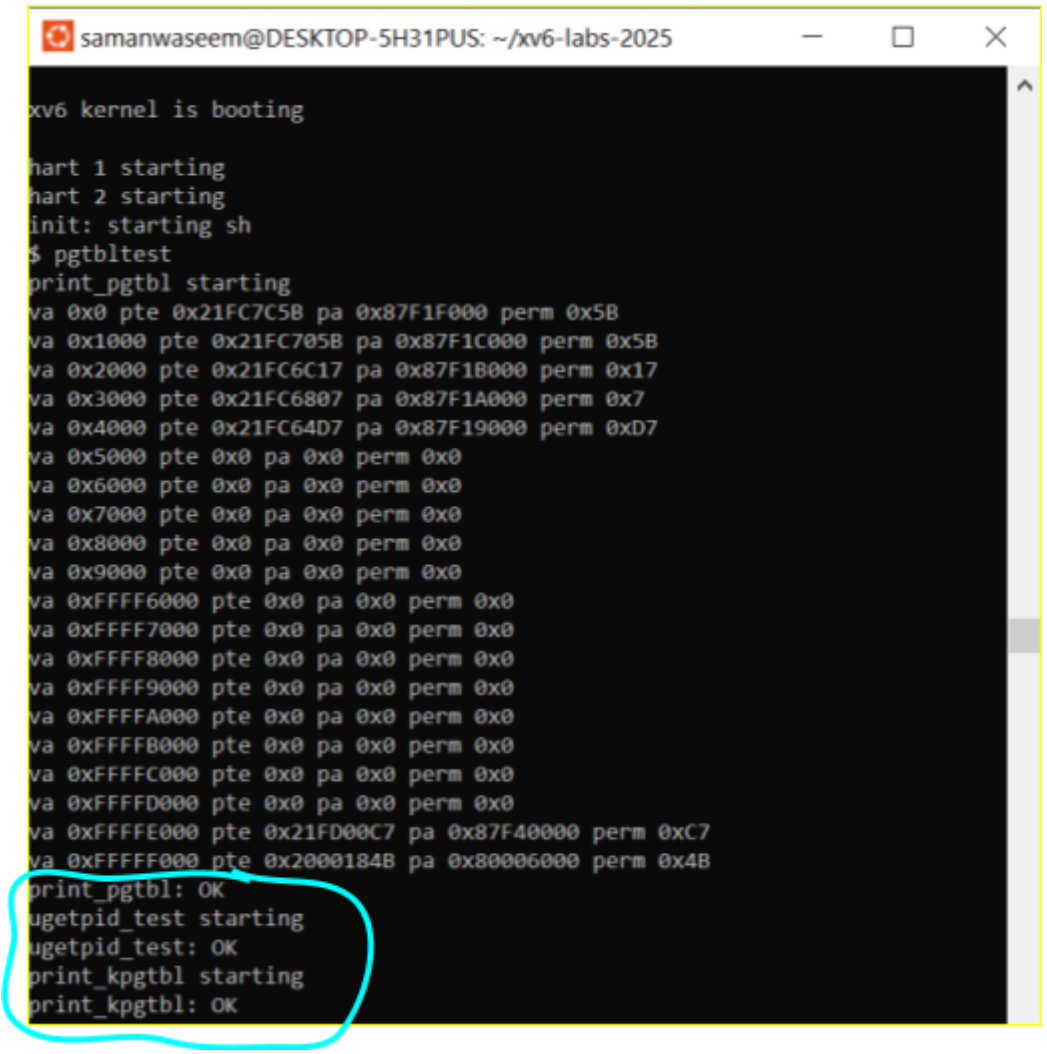


Figure 3.4: A process's user address space, with its initial stack.

2. Speed up system calls

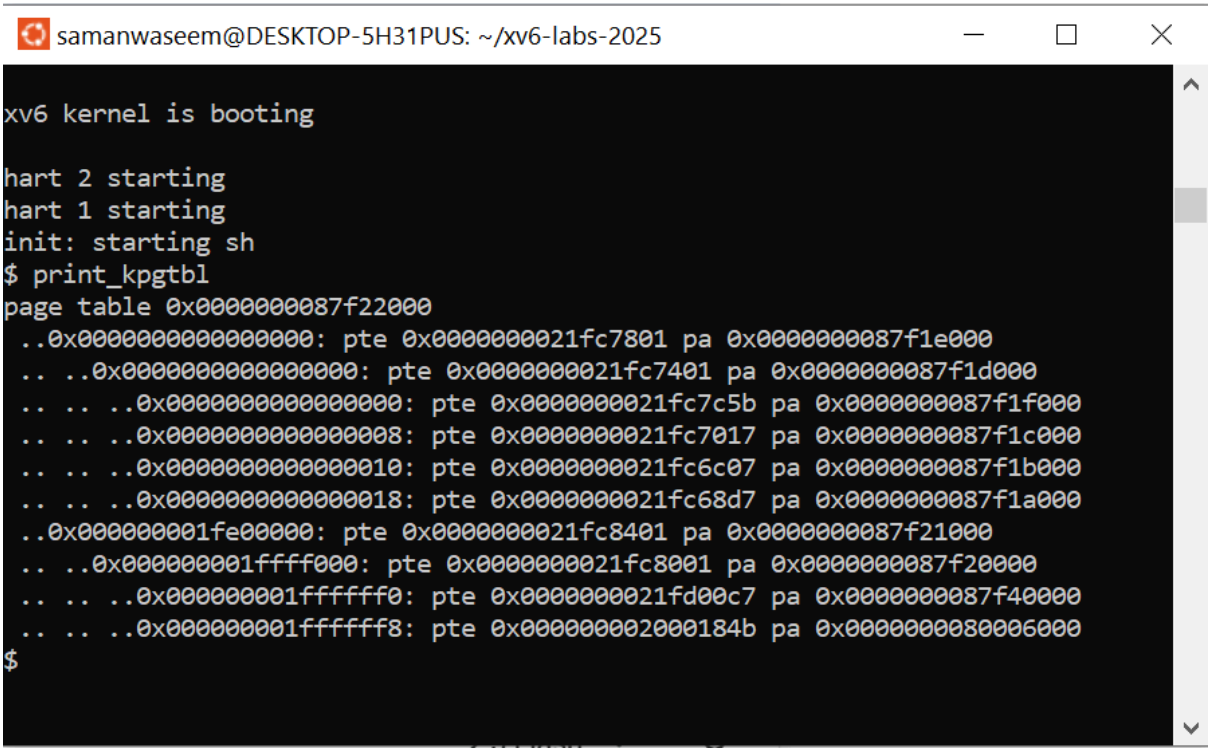


print_pgtbl: OK
ugetpid_test starting
ugetpid_test: OK
print_kpgtbl starting
print_kpgtbl: OK

Which other xv6 system call(s) could be made faster using this shared page? Explain how.

getpid(), which fetched parent processes' pids could be made much faster through a shared page because it will be able to avoid 'round trips' to the kernel through ecalls. By allocating a field in the shared page, the kernel would be able to only update it during rare events. When getpid() is called, the wrapper function would read the value directly. We won't need to switch between user and kernel mode.

3.Print a page table



‘page table 0x0000000087f22000’ shows a top level physical address
The indentation for the rest show levels 1 (. .) 2 (. . . .) and 3 (. . . . . .)
Permissions work the same way as shown in the analysis in Part 1.

Addresses are also distinguished as low or high addresses.